Question Papers

MCA - Sem - II

CAE - I + II + ESE.

CAE - I

[Number Of Question : 3] Seat No : [Total Number of Page :1]

G H raisoni COLLEGE | Engineering and Management Nagpur

C#.-INT-CAE-I-REG-1 SUMMER 2025
C#.NET (24MCAL202)

Time : ~~[3 Hours]~~ 90 min Max.Marks : [30.0]

Instruction to Candidate :
1) All Questions are compulsary.
2) Neat digram must be drawn wherever necessary.
3) Figures to the right indicate full marks.
4) Assume suitable data, if nececssay.
5) Use of non- programmable electronic scientific calculator is allowed.
6) Use of stream table,psychrometric chart,data is permitted.

Course Outcomes (CO) :
CO1-Describe the purpose of C#, and explain how to use Microsoft Visual C# and Visual Studio to build .NET Framework applications.
CO2-Describe the syntax of basic C# programming constructs.
CO3-Describe how to create and call methods, and explain Struct, Enum, Arrays and String.

Que.	Sub Que.		Marks	COs	BT Level
Q. 1	a	Explain the following: 1. Struct type 2. Enum type Write the application of C# use in current technology.	12.0	CO1	2
Q. 2	a	Differentiate in between Method overloading and Methos overriding in C#.Net. State the name of all the C# access modifiers.	12.0	CO2	4
Q. 3	a	Write a C# program with output to illustrate the enums with their default values.	6.0	CO3	1

[Number Of Question : 3] Seat No : [Total Number of Page :1]

GH raisoni COLLEGE | Engineering and Management Nagpur

CLO-INT-CAE-I-REG-1 SUMMER 2025
CLOUD COMPUTING (24MCAL201)

Time : [~~3 Hours~~ 90 min] **Max.Marks :** [30.0]

Instruction to Candidate :
1) All Questions are compulsary.
2) Neat digram must be drawn wherever necessary.
3) Figures to the right indicate full marks.
4) Assume suitable data, if nececssay.
5) Use of non- programmable electronic scientific calculator is allowed.
6) Use of stream table,psychrometric chart,data is permitted.

Course Outcomes (CO) :
CO1-Demonstrate the computing paradigms.
CO2-Ability to understand the ways in which the cloud fundamentals can be programmed and deployed.
CO3-Ability to understand various service delivery models of a cloud computing architecture and management of cloud.

Que.	Sub Que.		Marks	COs	BT Level
Q. 1	a	Explain the following Terms: 1. Cluster Computing 2. Quantum Computing	12.0	CO1	2
Q. 2	a	Describe 5-4-3 principles of Cloud Computing.	12.0	CO2	2
Q. 3	a	Explain cloud architecture with suitable diagram.	6.0	CO3	2

[Number Of Question : 3] Seat No : [Total Number of Page :1]

G H raisoni COLLEGE | Engineering and Management Nagpur

DAT-INT-CAE-I-REG-1 SUMMER 2025
DATA STRUCTURES (24MCAL204)

Time : [~~3 Hours~~ 90 min] Max.Marks : [30.0]

Instruction to Candidate :
1) All Questions are compulsary.
2) Neat digram must be drawn wherever necessary.
3) Figures to the right indicate full marks.
4) Assume suitable data, if nececssay.
5) Use of non- programmable electronic scientific calculator is allowed.
6) Use of stream table,psychrometric chart,data is permitted.

Course Outcomes (CO) :
CO1-1. Describe how arrays are represented in memory and various operations on it.
CO2-2. Describe representation of linked list, all possible operations on it.
CO3-3. Demonstrate stack, it representation, evaluation of expressions and recursion.

Que.	Sub Que.		Marks	COs	BT Level
Q. 1	a	What is Data Structure? Explain Array. Write an algorithm to insert an element in the middle of an array.	12.0	CO1	1
Q. 2	a	Define Linked List. Explain its types. Write an algorithm to traverse the entire linked list.	12.0	CO2	2
Q. 3	a	Draw the stack structure in each case when the following operations are performed on an empty stack. i) Add A, B, C, D, E. ii) Delete two letters iii)Add P	6.0	CO3	3

[Number Of Question : 3] Seat No : [Total Number of Page :1]

GH raisoni COLLEGE | Engineering and Management Nagpur

COM-INT-CAE-I-REG-1 SUMMER 2025
COMMUNICATION SKILLS (24MCAL210)

Time : [~~3 Hours~~ 90 min] **Max.Marks :** [15.0]

Instruction to Candidate :
1) All Questions are compulsary.
2) Neat digram must be drawn wherever necessary.
3) Figures to the right indicate full marks.
4) Assume suitable data, if nececssay.
5) Use of non- programmable electronic scientific calculator is allowed.
6) Use of stream table,psychrometric chart,data is permitted.

Course Outcomes (CO) :
CO1-To communicate effectively with coherence and relevance in speech and writing
CO2-To participate in brainstorming sessions for cooperative learning

Que.	Sub Que.		Marks	COs	BT Level
Q. 1	a	Explain the process of Communication	5.0	CO1	2
Q. 2	a	Write five differences between written and oral Communication	5.0	CO1	1
Q. 3	a	Write and explain five skills which are tested in Group Discussion	5.0	CO2	1

[Number Of Question : 3] Seat No : [Total Number of Page :1]

GH raisoni COLLEGE | Engineering and Management Nagpur

SYS-INT-CAE-I-REG-1 SUMMER 2025
SYSTEM ANALYSIS & DESIGN (24MCAL205)

Time : [~~3 Hours~~ 90 min] **Max.Marks :** [30.0]

Instruction to Candidate :
1) All Questions are compulsary.
2) Neat digram must be drawn wherever necessary.
3) Figures to the right indicate full marks.
4) Assume suitable data, if nececssay.
5) Use of non- programmable electronic scientific calculator is allowed.
6) Use of stream table,psychrometric chart,data is permitted.

Course Outcomes (CO) :
1-Explain the purpose and activities of the systems development life cycle phases.
2-To understand System planning and initial investigation.
3-Understand and model system processes, events, and data flows within a system.

Que.	*Sub Que.*		Marks	COs	BT Level
Q. 1	a	Explain the Feasibility Study.	12.0	1	2
Q. 2	a	Define the key objectives of the JAD session.	12.0	2	2
Q. 3	a	Explain the Agile Model.	6.0	3	2

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G H raisoni COLLEGE | Engineering and Management Nagpur

ORA-INT-CAE-I-REG-1 SUMMER 2025
ORACLE & PL/SQL (24MCAL203)

Time : [~~3 Hours~~ 90 min] **Max.Marks :** [30.0]

Instruction to Candidate :
1) All Questions are compulsary.
2) Neat digram must be drawn wherever necessary.
3) Figures to the right indicate full marks.
4) Assume suitable data, if nececssay.
5) Use of non- programmable electronic scientific calculator is allowed.
6) Use of stream table,psychrometric chart,data is permitted.

Course Outcomes (CO) :
CO1-Understanding basic database security and object privileges.
CO2-Using the powerful set of built-in SQL functions.
CO3-Protecting the database and extending the basic data model using declarative constraints.

Que.	Sub Que.		Marks	COs	BT Level
Q. 1	a	Explain the Oracle Architecture in detail.	12.0	CO1	2
Q. 2	a	What is Physical Backup and Logical Backup? Also state the difference between them.	12.0	CO2	1
Q. 3	a	Explain Arithmetic Operator of SQL.	6.0	CO3	2

CAE - II

[Number Of Question : 3]

Seat No :

[Total Number of Page :1]

G H raisoni COLLEGE | Engineering a Management Nagpur

C#.-INT-CAE-II-REG-1 SUMMER 2025
C#.NET (24MCAL202)

Question Set Name : CAE-II

Time : 1 Hr 30 Mins

Max.Marks : [30]

Course Outcomes (CO) :
CO3-Describe how to create and call methods, and explain Struct, Enum, Arrays and String.
CO4-Describe how to catch, handle, and throw exceptions.
CO5-Describe how to perform basic file I/O operations in a Visual C# application C.

Que	Sub Que	Question	Marks	COs	bt leval
1	(a)	Describe with examples Method used in ArrayList in C#.	(6)	CO3	2
2	(a)	Define Web Services?	(2)	CO4	1
	(b)	Which are the technologies used in Web Services?	(10)	CO4	1
3	(a)	Define Managed Code in .Net with diagram.	(6)	CO5	1
	(b)	Define Unmanaged Code in .Net with diagram.	(6)	CO5	1

[Number Of Question : 3] Seat No : [Total Number of Page :1]

G H raisoni COLLEGE | Engineering a Management Nagpur

CLO-INT-CAE-II-REG-1 SUMMER 2025
CLOUD COMPUTING (24MCAL201)

Question Set Name : CAE-II

Time : 1 Hr 30 Mins **Max.Marks : [30]**

Course Outcomes (CO) :
CO3-Ability to understand various service delivery models of a cloud computing architecture and management of cloud.
CO4-Ability to understand the cloud service Modelconcepts like Iaas, Saas, Paas.
CO5-Understanding cloud service providers.

Que	Sub Que	Question	Marks	COs	bt leval
1	(a)	What are the unique properties of Cloud Applications?	(6)	CO3	1
2	(a)	Explain in detail Infrastructure as a Service (IaaS).	(6)	CO4	2
	(b)	Define Platform as a Service Model. Also write its Pros and Cons in detail.	(6)	CO4	1
3	(a)	Write notes on Amazon Simple Storage Service and Amazon Simple Queue Service.	(12)	CO5	1

[Number Of Question : 3] Seat No : [Total Number of Page :1]

G H raisoni COLLEGE | Engineering a Management Nagpur

COM-INT-CAE-II-REG-1 SUMMER 2025
COMMUNICATION SKILLS (24MCAL210)

Question Set Name : CAE-II

Time : 1 Hr 30 Mins **Max.Marks : [15]**

Course Outcomes (CO) :
CO2-To participate in brainstorming sessions for cooperative learning
CO3-To expose the learners to cultural diversity and value education through the humanistic curriculum

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Answer any **3** of following questions

Que	Sub Que	Question	Marks	COs	bt leval
		describe			
		Section : Answer all questions			
1		Write the importance of Organisational with examples?	(5)	CO2	2
2		Write 5 strategies that can enhance the possibility of success in an Interview?	(5)	CO2	3
3		What are the essential skills of a seasoned Negotiator?	(5)	CO3	1

[Number Of Question : 3]

Seat No :

[Total Number of Page :1]

G H raisoni COLLEGE | Engineering a Management Nagpur

DAT-INT-CAE-II-REG-1 SUMMER 2025
DATA STRUCTURES (24MCAL204)

Question Set Name : CAE-II

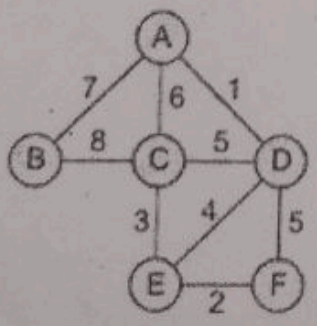
Time : 1 Hr 30 Mins

Max.Marks : [30]

Course Outcomes (CO) :
CO3-3. Demonstrate stack, it representation, evaluation of expressions and recursion.
CO4-4. Describe queues, operations using array and linked list, searching and sorting.
CO5-5. Demonstrate different methods for traversing trees and graph

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Que	Sub Que	Question	Marks	COs	bt leval
		1			
1	(a)	Define the term Prefix & Postfix Expression. Convert A + B*C-D/ E into postfix equivalents.	(6)	CO3	03
2	(a)	Compare Selection sort and Insertion Sort algorithm with an example.	(8)	CO4	03
	(b)	What are the advantages of using a circular queue over a linear queue?	(4)	CO4	02
3	(a)	Construct a minimum spanning tree of the graph using Kruskal's algorithm.	(6)	CO5	04



	(b)	Define the following with example i) Graph ii) Degree of a node iii) Closed Path iv) Cycle v) Complete Graph	(6)	CO5	01

[Number Of Question : 3] Seat No : [Total Number of Page :1]

G H raisoni COLLEGE | Engineering a Management Nagpur

SYS-INT-CAE-II-REG-1 SUMMER 2025
SYSTEM ANALYSIS & DESIGN (24MCAL205)

Question Set Name: CAE-II

Time : 1 Hr 30 Mins **Max.Marks :** [30]

Course Outcomes (CO) :
3-Understand and model system processes, events, and data flows within a system.
4-To understand the concept of Data Dictionary
5-To understand hardware /software selection

Compulsory

Que	Sub Que	Question	Marks	COs	bt leval
1	(a)	What is the difference between a logical and physical data flow diagram?	(6)	3	1
2	(a)	What is a data dictionary and explain its purpose in a database system	(12)	4	1
3	(a)	Explain in detail the cost/benefit analysis.	(12)	5	2

ESE - Sem - II

END SEM EXAMINATION

SUMMER 2025

GH raisoni COLLEGE
Engineering and Management Nagpur

Name of Program:	Master in Computer Application	Sem:	2nd
Name of Course :	ORACLE & PLSQL	Course Code:	24MCAL203
Max Marks:	50 Marks	Duration:	180 Mins.

INSTRUCTION TO THE STUDENTS

1. *Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.*
2. *Solve **Q. 4 OR Q. 5** remaining questions are mandatory*
3. *Every question has equal weightage.*
4. *Use of programmable calculator is prohibited.*
5. *Assume suitable data wherever necessary.*
6. *Draw neat and proper diagram/sketches.*
7. *Don't use red pen for writing the answers.*
8. *Don't write any other comments except answers of questions.*

ABBREVIATIONS

Q.: Question Number S.Q.: Sub Question Number BT: Blooms taxonomy Level CO: Course Outcome

LIST OF COURSE OUTCOME

CO1: Understanding basic database security and object privileges.
CO2: Using the powerful set of built-in SQL functions.
CO3: Protecting the database and extending the basic data model using declarative constraints.
CO4: Performing advanced SQL queries such as grouping and Join, Sub queries, PL/SQL Blocks, functions.
CO5: Use the PL/SQL code constructs of IF-THEN-ELSE and LOOP types as well as syntax and command function

Q.	S.Q.	Question	Marks	BT Level	CO
1	a	Discuss in detail the architecture of ORACLE Database.	10	2	CO1
2	a	What is backup? Explain the concept of Oracle backup and recovery configuration.	10	1	CO2
3	a	Explain Arithmetic function of SQL.	05	2	CO3
	b	Explain the Integrity constraint in detail.	05	2	CO3
4	a	Write about Sequences with Syntax and Parameters.	05	1	CO4
	b	Discuss about Views in detail.	05	2	CO4
		OR			
5	a	What is sub query? What if sub query returns more than one value? Write a query to select all the fields of the first table SALARY and the names of employees from the second table EMP for those employees who had less than average salary.	10	1	CO4
6	a	Write short note on Triggers. Explain in Detail.	10	1	CO5

END SEM EXAMINATION
SUMMER 2025

GHraisoni COLLEGE
Engineering and Management
Nagpur

Name of Program:	Master in Computer Application	Sem:	2nd
Name of Course :	C#.NET	Course Code	24MCAL202
Max Marks:	50 Marks	Duration:	180 Mins.

INSTRUCTION TO THE STUDENTS

1. *Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.*
2. *Solve **Q. 3 OR Q. 4**, remaining questions are mandatory*
3. *Every question has equal weightage.*
4. *Use of programmable calculator is prohibited.*
5. *Assume suitable data wherever necessary.*
6. *Draw neat and proper diagram/sketches.*
7. *Don't use red pen for writing the answers.*
8. *Don't write any other comments except answers of questions.*

ABBREVIATIONS

Q.: Question Number S.Q.: Sub Question Number BT: Blooms taxonomy Level CO: Course Outcome

LIST OF COURSE OUTCOME

CO1: Describe the purpose of C#, and explain how to use Microsoft Visual C# and Visual Studio to build .NET Framework applications.
CO2: Describe the syntax of basic C# programming constructs.
CO3: Describe how to create and call methods, and explain Struct, Enum, Arrays and String
CO4: Describe how to catch, handle, and throw exceptions
CO5: Describe how to perform basic file I/O operations in a Visual C# application C

Q.	S.Q.	Question	Marks	BT Level	CO
1	a	Describe the general structure of C# Program.	05	01	CO1
	b	Explain the following: a. Value Type b. Reference Type	05	02	CO1
2	a	Discuss how a class is declared and object is accessed to the class member.	04	03	CO2
	b	Explain the class member in C# with examples.	06	02	CO2
3	a	Write a correct way to settle down values into the structure variable 'e' defined in the following C# code snippet?		03	CO3

```
struct emp
{
public String name;
public int age;
public Single sal;
}
```

04

```
emp e = new emp();
```

	b	What is struct in C#.Net? Explain with an example?	06	02	CO3

OR

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END SEM EXAMINATION

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Q.	S.Q.	Question	Marks	BT Level	CO
	a	Describe Array in C#.	02	01	CO3
	b	Explain the various way to initialize array in C#.	03	03	CO3
	c	Write a program in C# to print 11 22 33 44 55 as an output using Array.	05	03	CO3
5	a	What is .Net framework?	02	01	CO4
	b	Explain its component in detail.	08	02	CO4
	a	Explain pinned object in C#.	10	02	CO5

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END SEM EXAMINATION

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SUMMER 2025

Name of Program:	**Master of Computer Application**	**Sem:**	**2nd**
Name of Course :	**Data Structures**	**Course Code**	**24MCAL204**
Max Marks:	**50 Marks**	**Duration:**	**180 Mins.**

INSTRUCTION TO THE STUDENTS

1. *Read the question paper carefully(Branch,Semester,Scheme)before attempting the questions.*
2. *Solve **Q. 1 OR Q. 2**, remaining questions are mandatory*
3. *Every question has equal weightage.*
4. *Use of programmable calculator is prohibited.*
5. *Assume suitable data wherever necessary.*
6. *Draw neat and proper diagram/sketches.*
7. *Don't use red pen for writing the answers.*
8. *Don't write any other comments except answers of questions.*

ABBREVIATIONS

Q.: Question Number S.Q.: Sub Question Number BT: Blooms taxonomy Level CO: Course Outcome

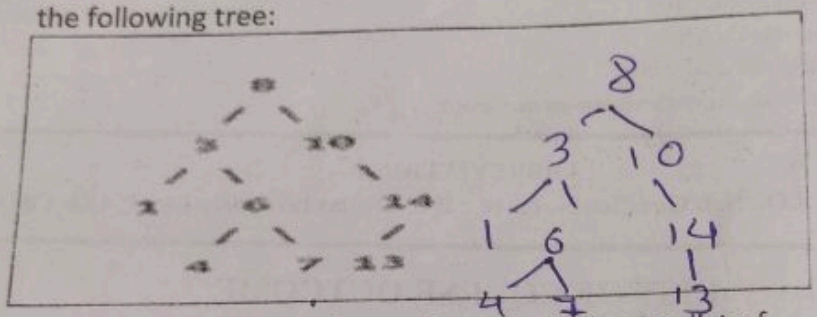
LIST OF COURSE OUTCOME

CO1: Describe how arrays are represented in memory and various operations on it.
CO2: Describe representation of linked list, all possible operations on it.
CO3: Demonstrate stack, its representation, evaluation of expressions and recursion.
CO4: Describe queues, operations using array and linked list, searching and sorting.
CO5: Demonstrate different methods for traversing trees and graph

Q.	S.Q.	Question	Marks	BT Level	CO
1	a	Explain array with example.	05	02	CO1
	b	Given an array int marks[]={99,67,78,56,88,90,34,85}, calculate the address of marks[4] if the base address = 1000.	05	03	CO1
		OR			
2		Data [] is an array that is declared as int Data [20]; and contains the following values: Data [] = {12, 23, 34, 45, 56, 67, 78, 89, 90, 100};			
	a	Calculate the length of the array.	02	03	CO1
	b	Find the upper_bound and lower_bound	02	04	CO1
	c	Show the memory representation of the array.	02	03	CO1
	d	If a new data element with the value 75 has to be inserted, find its position.	02	04	CO1
	e	Insert a new data element 75 and show the memory representation after the insertion.	02	04	CO1
3	a	What are the advantages of linked list over arrays?	04	02	CO2
	b	Write algorithm for inserting and deleting a node at the beginning from a singly linked list.	06	03	CO2

END SEM EXAMINATION

GHraisoni COLLEGE
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Q.	S.Q.	Question	Marks	BT Level	CO
4	a	How is a stack represented in memory using an array? Explain with an example.	05	04	CO3
	b	Define the term prefix expression. Convert A + B * C – D / E into postfix equivalents.	05	03	CO3
5	a	How is a queue represented in memory using an array?	05	04	CO4
	b	Describe the insertion sort techniques with an example.	05	03	CO4
6	a	Perform **In-order, Pre-order and post-order** traversal on the following tree: 	06	03	CO5
	b	Construct a binary search tree from the following list of elements inserted in order: **[10, 5, 15, 2, 7, 12, 18,24,27,3,11,16]**. Show the tree representation.	04	04	CO5

END SEM EXAMINATION

SUMMER 2025

GHraisoni COLLEGE
Engineering and Management Nagpur

Name of Program:	Master in Computer Application	Sem:	2nd
Name of Course :	System Analysis and Design	Course Code:	24MCAL205
Max Marks:	50 Marks	Duration:	180 Mins.

INSTRUCTION TO THE STUDENTS

1. *Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.*
2. *Solve **Q. 2 OR Q. 3** remaining questions are mandatory*
3. *Every question has equal weightage.*
4. *Use of programmable calculator is prohibited.*
5. *Assume suitable data wherever necessary.*
6. *Draw neat and proper diagram/sketches.*
7. *Don't use red pen for writing the answers.*
8. *Don't write any other comments except answers of questions.*

ABBREVIATIONS

Q.: Question Number S.Q.: Sub Question Number BT: Blooms taxonomy Level CO: Course Outcome

LIST OF COURSE OUTCOME

CO1: Explain the purpose and activities of the systems development life cycle phases.
CO2: To understand System planning and initial investigation.
CO3: Understand and model system processes, events, and data flows within a system.
CO4: *To understand the concept of Data Dictionary.*
CO5: *To understand hardware /software selection.*

Q.	S.Q.	Question	Marks	BT Level	CO
1	a	List and define the seven phases of the systems development life cycle (SDLC).	05	1	CO1
	b	What are CASE tools? Explain the uses of it.	05	1	CO1
2	a	Define what is meant by closed interview questions. Give six benefits and four drawbacks of using them.	05	1	CO2
	b	When are open-ended questions appropriate for use in interviewing?	05	1	CO2
		OR			
3	a	What information about the decision maker does the analyst seek to gain from observation?	06	1	CO2
	b	List the three approaches to complex random sampling.	04	1	CO2

END SEM EXAMINATION

GHraisoni COLLEGE
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Q.	S.Q.	Question	Marks	BT Level	CO
4	a	What is meant by the term patched-up prototype?	05	1	CO3
	b	What are the four core practices of the agile approach?	05	1	CO3
5	a	List three reasons for creating a logical data flow diagram.	06	1	CO4
	b	Illustrate when transaction files required in the system design.	04	3	CO4
6	a	Explain in detail cost/benefit analysis.	10	2	CO5